$$\frac{d^2E}{d\omega d\theta} = \frac{h^2\eta^2}{4\pi c} \frac{A\sigma^2 \left|\Phi\left(\omega\right)\right|^2 \left(\varepsilon - 1\right)^2 \left[\left(1 - \beta^2\right)\varepsilon + \beta^2 \cos^2\theta_0\right]}{\left(\sqrt{1 - \varepsilon\beta^2} + \varepsilon\sqrt{1 - \beta^2}\right)^2 \left(\sqrt{\varepsilon - \cos^2\theta_0} + \varepsilon \sin\theta_0\right)^2} \delta\left(\theta - \theta_0\right)$$

$$\sigma = \int_{-\infty}^{+\infty} \mathrm{d}x\mathrm{d}z\, n\left(x, z\right), \quad \Phi\left(\omega\right) = \frac{1}{\sigma} \int_{-\infty}^{+\infty} \mathrm{d}x\mathrm{d}z\, e^{-i\varkappa x} e^{-\eta|z|} n\left(x, z\right)$$

$$\varkappa = \frac{\omega}{c\beta}, \quad \eta = \frac{\omega}{c}\sqrt{1/\beta^2 - \varepsilon}, \quad \cos\theta_0 = \frac{1}{\beta} - \frac{2\pi c}{a\omega}, \quad \beta = \frac{v}{c}$$